When Disagreement Labels Do Not Identify F1 Rankings

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Abstract. When two binary classifiers agree, their labels cancel from an accuracy comparison but can still change an F1 ranking. We specialize partial identification to an explicit audit: after labeling all disagreements, when can one choose a model, and when must agreements be queried? The criterion certifies either an exact ranking or a choice within an F1 tolerance of the better model for every compatible completion. Across 3,200 frozen model pairs, retaining the shared completion tightens bounds and a 0.01 tolerance substantially reduces labeling costs. A zero-loss (weak-winner) control shows that this saving comes mainly from tolerating a small positive loss, rather than merely permitting ties. The guarantee concerns the evaluated batch; it does not estimate performance on future data.

1 What can disagreement labels decide?

Suppose two fixed classifiers disagree on three of ten items. One predicts positive on all three, the other negative; only one of these labels is positive. They agree positive on three other items and negative on four. Even knowing that exactly four of the ten labels are positive does not settle their F1 ranking. Allocating the three remaining positives to negative agreements gives F1 scores $(1/5, 0)$; allocating them to positive agreements gives $(4/5, 6/7)$. The same disagreement labels and prevalence support opposite winners.

Given two prediction vectors, their disagreement labels and optional prevalence, our audit reduces unknown agreement labels to two counts. The resulting exact F1 interval either certifies a choice at the requested tolerance or requires more agreement labels. This explicit solution makes the stopping rule directly computable while preserving a *shared* label completion.

Discordant cases also determine McNemar's paired error-rate statistic [1]. That is a population hypothesis test; we identify what fixed-batch F1 comparisons permit under every compatible labeling.

Active evaluation estimates F-measures [2], compares expected losses [3], estimates risk without bias [4], or uses Bayesian equivalence and consensus priors [5, 6]. Our guarantee covers every compatible completion. F1 batch dependence [7] and missing-label bounds [8] are established. We specialize the target-domain certificate framework (TDC) [9] and Polo et al. [10] to an explicit audit.

2 An explicit audit criterion

Fix binary predictions $q, c \in \{0,1\}^n$ and correct unknown labels. Let a, b count disagreements $(q,c) = (1,0), (0,1)$; n_{00}, n_{11} count agreements. After all $D = a+b$ disagreement labels are revealed, let u, v be their respective positive counts. Unknown positive agreement counts are s, t. Write

$$P = u + v + s + t, \quad Q = a + n_{11}, \quad C = b + n_{11}, \quad d = u - v.$$

P is the true positive count, while Q, C are observable predicted counts. Positive-class F1 is $2\mathrm{TP}/(P + \text{predicted positives})$, with $0/0 = 0$. The paired difference is

$$\Delta(s,t) = F_1(q) - F_1(c) = \frac{2(t+u)}{P+Q} - \frac{2(t+v)}{P+C}. \tag{1}$$

By contrast, accuracy difference $(2d + b - a)/n$ is already determined; balanced accuracy difference $[nd + (b-a)P]/[2P(n-P)]$ is determined if $0 < P < n$ is known, and undefined at $P = 0, n$.

Sharp sign criterion (specialization). Set $e = d + b - a$, $K = d(u + v + n_{11}) + ub - va$. Then Δ has the sign of

$$S(s,t) = K + ds + et. \tag{2}$$

Without prevalence information, agreement queries leave an integer rectangle $s \in [s_-, s_+]$, $t \in [t_-, t_+]$, initially $[0, n_{00}] \times [0, n_{11}]$. Its exact numerator bounds are

$$\begin{aligned} S_- &= K + \min(ds_-, ds_+) + \min(et_-, et_+), \\ S_+ &= K + \max(ds_-, ds_+) + \max(et_-, et_+). \end{aligned} \tag{3}$$

Both strict rankings are possible iff $S_- < 0 < S_+$. A strict winner is identified when $S_- > 0$ or $S_+ < 0$; an exact tie when $S_- = S_+ = 0$. Remaining cases permit a tie and one winner.

Proof. Cross multiplication gives $\Delta(P+Q)(P+C)/2 = dP + (C-Q)t + uC - vQ = K + ds + et$. The denominator is positive unless $P = 0$ and a predictor has no positives; then both F1 scores and S are zero. Every integer count in the rectangle is attainable. An affine function reaches its extrema at its corners, proving the sign claims. □

Equal-count corollary. If $Q = C = R > 0$, then $\Delta = 2d/(P+R)$: disagreement labels identify the sign without knowing P. If $R = 0$, both scores vanish. Magnitude can remain unknown.

Exact magnitude bounds. For fixed $P > 0$, the difference is affine in t:

$$f_P(t) = \frac{2u}{P+Q} - \frac{2v}{P+C} + \frac{2(C-Q)t}{(P+Q)(P+C)}. \tag{4}$$

The feasible endpoints are $\ell_P = \max(t_-, P - u - v - s_+)$ and $h_P = \min(t_+, P - u - v - s_-)$. If $\ell_P \le h_P$, evaluating f_P at both endpoints gives its exact hull;

otherwise that P is infeasible. At $P = 0$ both scores are zero. With unknown P, enumerate feasible integer counts P and take the smallest and largest of these endpoints, yielding the exact interval hull $[L, U]$ in $O(n)$ time. This includes $P = n$. The sign proof does not place unknown-P magnitude extrema at rectangle corners.

A choice with bounded regret. For absolute F1 tolerance $\varepsilon \geq 0$, the worst losses from selecting q and c are $R_q = \max(0, -L)$ and $R_c = \max(0, U)$. Choose the smaller, breaking ties in favor of q, and stop iff

$$\min(R_q, R_c) \leq \varepsilon \quad \Longleftrightarrow \quad L \geq -\varepsilon \text{ or } U \leq \varepsilon. \tag{5}$$

For every compatible completion, choosing q loses at most $\max(0, -L)$ relative to the better model; the analogous bound for c is $\max(0, U)$. Thus (5) guarantees an ε-best choice on this batch. It does not require $[L, U] \subseteq [-\varepsilon, \varepsilon]$. At $\varepsilon = 0$ it allows a weak winner before distinguishing a strict win from an exact tie. Failure means $L < -\varepsilon$ and $U > \varepsilon$: opposite compatible advantages exceeding the tolerance. The original material-reversal definition $L \leq -\varepsilon$, $U \geq \varepsilon$ differs only at the boundaries; the regret criterion is the decision interpretation of this condition.

Relation to general identification frameworks. TDC [9], Theorem 1, gives sharp certificates for fixed additive-loss contrasts over unrestricted completions. With $P > 0$ fixed,

$$\Delta = \sum_i y_i \left[\frac{2q_i}{P + Q} - \frac{2c_i}{P + C} \right]. \tag{6}$$

Imposing $\sum_i y_i = P$ is essential for sharpness; TDC's balanced-accuracy construction (§4.5) already imposes such a constraint and enumerates prevalence. Equations (2)–(4) are explicit specializations.

Polo et al. [10], Eq. (1.1) and §3.2, also cover fixed-P F1: use uniform batch indices, distinguish audited items and fix the label marginal. For (6), the remaining fractional labels have an integer sum. Every vertex is integral: two fractional entries permit a sum-preserving perturbation; one cannot sum to an integer. Hence linear extrema equal our completion extrema. Our contribution is the explicit audit and its consequences.

Separate bounds form $[\min F_1(q) - \max F_1(c), \max F_1(q) - \min F_1(c)]$. Every extremum uses the *same* feasible completions and prevalence constraints, but its optimizers may be incompatible. Fixed-P marginals are affine in t, so the same enumeration computes these conservative bounds exactly.

3 Retrospective evaluation

Frozen cases and units. We reuse TDC's prediction artifact, release 1.1.6 [11], without refitting models. Each of 36 cells contains 500 targets, a validation-selected quantum-kernel predictor, and 115 classical configurations. Twenty cells cover BRFSS diabetes and ACS FoodStamps (five subsampling seeds, SVC and

Gaussian-process classification); 16 cover eight EMBER, ToN-IoT and UNSW-NB15 scenarios with both classifiers. Deduplicating identical classical predictions within cells yields 3,200 pairs. The criterion applies to arbitrary fixed binary predictions.

The eight-cell pilot (801 pairs) was fixed before reading labels in this study. The 28 reserved cells (2,399 pairs) are an analytical holdout of existing evidence. Tolerance, separate-bound and validation-comparator analyses are retrospective extensions, not prospective confirmation.

Fractions average equally across cells; costs average orders, pairs within cells, then eligible cells. Shared targets and overlapping subsamples preclude independent-observation inference; summaries are descriptive. Known P is an exact information oracle. Initial intervals retain $0 < P < n$, the original balanced-accuracy domain; query cohorts and stops allow $P = 0, n$.

Acquisition and comparators. We retain three agreement orders: uniform; alternating shuffled agreement strata; and decreasing $|d|$ or $|e|$ (magnitude, shuffled ties). Seeds 0–31 share permutations across policies and objectives. Orders use predictions and disagreement labels only; queries shrink the count intervals. Magnitude minimizes numerator width, not necessarily cost. We compare strict-sign/exact-tie identification, a weak winner ($\varepsilon = 0$), and $\varepsilon = 0.01$ on the same trajectories. Full labels verify regret; they never select pairs, queries or stopping decisions.

We also compare one classical predictor per cell: the closest in Hamming distance (original-order tie break), and the maximum validation balanced-accuracy candidate (lexicographic configuration tie break). Validation scores cover all 115 candidates; choices preceded the new target analysis. This retrospective rule follows TDC's objective, not validation F1. Hamming proximity is a sensitivity analysis, not a deployment rule.

Mechanism. At known $P > 0$, interval width is exactly $2|C-Q|(h_P-\ell_P)/[(P+Q)(P+C)]$: predicted-count imbalance and freedom to allocate positives between agreement groups jointly drive ambiguity. Without P, initial numerator width is $|d|n_{00}+|e|n_{11}$. All 58 equal-count pairs identify the sign, although their mean magnitude interval remains nonzero. Disagreement count alone therefore does not describe this uncertainty.

Initial decisions. In the 28 reserved cells, the mean material-reversal fraction is 20.54% without P, and 10.52% with exact P. These count pairs admitting compatible winners in both directions with margins at least 0.01, not observed mistakes or a universal frequency. No pair lies on the relevant 0.01 boundaries, so the complements are certifiable choices at that tolerance.

Keeping dependence matters: in the reserved cells, paired bounds certify 79.46% of choices at $\varepsilon = 0.01$, versus 6.57% for separate bounds; with exact P, 89.48% versus 18.80%. Mean widths are respectively 0.173 versus 1.585, and 0.034 versus 1.005. This is an information ablation on identical feasible sets, not a comparison with jointly applied TDC or Polo.

Figure 1 shows the tolerance dependence: at 0.01, certification is 79.15% over all 36 cells, versus 79.46% in the 28 reserved cells. Hamming selection leaves material ambiguity in **3/36** cells; validation selection leaves **6/36** (both have 11/36 strict reversals). These one-pair-per-cell designs answer a different question from the all-configuration pool.

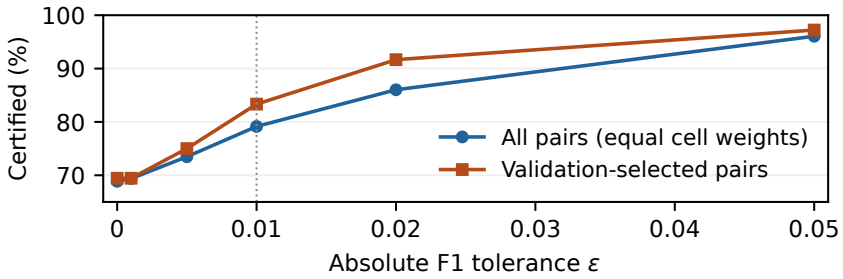


Fig. 1: Certification after disagreement labeling, unknown prevalence ($0 < P < n$). Equal-cell means over all 36 cells: 3,200 pairs, or one validation-selected pair per cell. Points are evaluated tolerances; lines guide the eye. Descriptive fractions, not confidence estimates.

Label costs. Table 1 uses common cohorts for both goals, including pairs already decided. Under magnitude ordering, replacing the exact goal by tolerance reduces total cost from 121.16 to 74.59 labels across all pairs and from 92.57 to 40.29 under validation selection, versus 500 for full labeling. The magnitude zero-loss control gives 120.92 (all pairs) and 92.25 (validation), close to strict/exact costs. Almost all the saving at 0.01 comes from permitting positive loss, rather than ties.

		Initial	Additional		Total	
Cohort	Policy	D	Exact	ε	Exact	ε
All pairs	Uniform	54.5	88.21	35.58	142.70	90.07
All pairs	Alternating	54.5	80.64	31.29	135.13	85.78
All pairs	Magnitude	54.5	66.67	20.10	121.16	74.59
Validation	Uniform	30.7	89.07	25.56	119.79	56.28
Validation	Alternating	30.7	77.58	16.34	108.30	47.07
Validation	Magnitude	30.7	61.84	9.57	92.57	40.29

Table 1: Mean labels on common cohorts, 36 cells, 32 orders per policy. All pairs: 3,200 comparisons; validation: one per cell. Additional labels query agreements; $\varepsilon = 0.01$. Each batch has 500 items. Totals are calculated before rounding.

Costs vary across cells: under magnitude ordering and $\varepsilon = 0.01$, the all-pair cell means have minimum/median/maximum 14.0/77.0/188.4 labels. This

is descriptive dispersion, not an uncertainty interval. The overall mean is 74.59 with equal cell weights, versus 82.05 with equal pair weights. In 15,466 of 307,200 dependent replays, the certified choice is the realized-worse model; maximum regret is 0.00927, within tolerance. The guarantee follows from (5), rather than from these retrospective checks.

4 Use and limitations

Given a tolerance, an auditor computes $[L, U]$ from the current count bounds, then chooses by (5) or queries agreements and updates those bounds.

This specialization assumes fixed predictions and correct labels; its guarantee does not extend to populations, noisy labels, macro-F1, multiclass or families. Compatible extremes need not be probable; benchmark reuse limits breadth. Disagreement-first labeling is imposed. Costs are neither optimal-acquisition lower bounds nor additive across pairs. Statistical alternatives can trade assumptions or error probability for fewer labels.

Availability. Code, inputs and results (version 1.1.0):
`https://doi.org/10.5281/zenodo.23647323`.

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